BRAUDE
College of Engineering, karmiel

Sequential Eye-Tracking Analysis System for Dynamic Behavioral Pattern Identification in ASD

CODE: 26-2-D-1

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Introduction: ASD & Eye Tracking



Research Context

This project is conducted in collaboration with the University of Haifa, as part of an eye-tracking experiment with children with ASD.

ASD can affect social communication and visual attention, so analyzing gaze during natural play can help reveal meaningful behavioral patterns.

Why Eye Tracking?

Eye tracking captures where and how children look, providing objective data about visual attention in a natural environment.

Atypical Visual Patterns in ASD

Circumscribed Attention

Focus on non-social details rather than social cues

Sticky Gaze

Difficulty shifting gaze away from one object

Eccentric Viewing

Looking at objects from unusual angles or side vision





Experimental Setup & Data Collection

Free Play Session

- Each child enters a dedicated playroom
- 10 minutes of natural free play
- No direct task instructions

Tobii Eye-Tracking Glasses

- Record the child's point-of-view video
- Capture continuous gaze coordinates

External Room Cameras

- Record the child's physical interactions and location in the room

The Current Challenge: Manual AOI Annotation

Areas of Interest (AOI's) in the Playroom

- Slide and a rubber tube
- Table with games
- Metal spring on a cushion

How AOI's Are Defined

Manual Marking
They manually draw and label relevant objects in each image.

Frame-by-Frame
In dynamic play, these areas must be adjusted for every single frame.

Static Snapshots
Researchers rely on static snapshots of the playroom scene.

Why Is It a Problem?

Time-Consuming
Requires long manual annotation for each session.

Highly Manual
Subjective, repetitive, and error-prone.

Difficult to Scale
Limits analysis of large datasets.

Project Objectives

Automated Dynamic AOI Analysis

Use neural networks to automatically detect and track objects in the playroom, replacing manual AOI annotation.

KEY OUTPUT

Identify which AOI the child is looking at over time.

Multimodal Behavioral Understanding

After reliable AOI detection, synchronize gaze data with external room cameras.

KEY OUTPUT

Understand whether the child is looking at the same object they are physically playing with.

Our Solution Steps

Step 01

Tobii Data Export

POV video + gaze coordinates.

Step 02

VLM-GIST AOI Detection

Automatically detects and tracks playroom objects.

Step 03

Gaze-to-AOI Mapping

Connects gaze points to the detected AOIs.

Step 04

Video Integration

Synchronizes POV gaze behavior with room-camera footage.

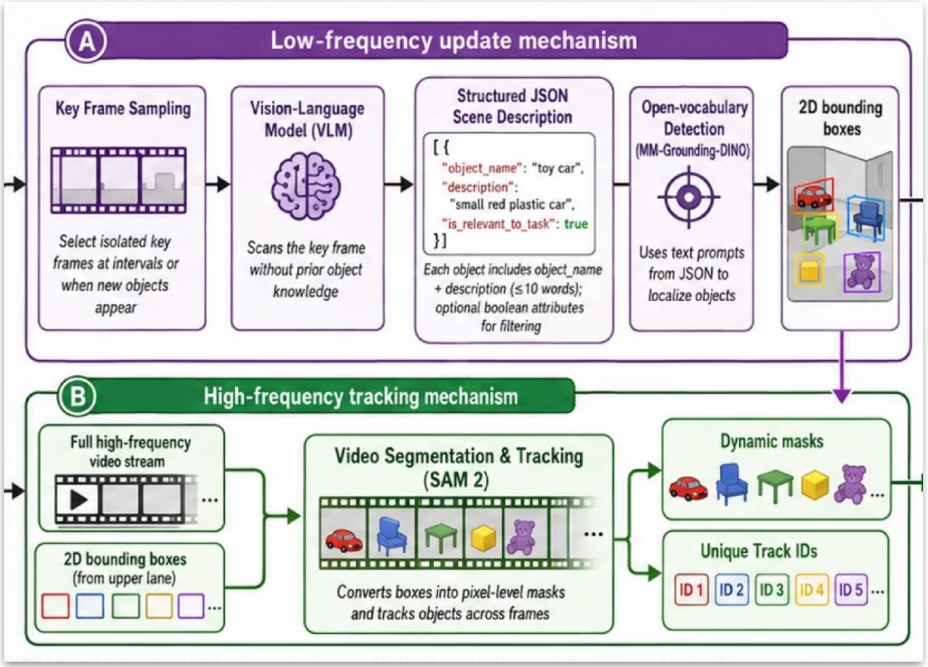
Output

Behavioral Insights

From raw recordings to meaningful behavioral insight.

VLM-GIST: Automated AOI Extraction

VLM-GIST **(VLM Grounding for Instance Segmentation & Tracking)** automates object detection and tracking without manual annotation, converting visual data into precise-tracking object masks through a two-phase pipeline:



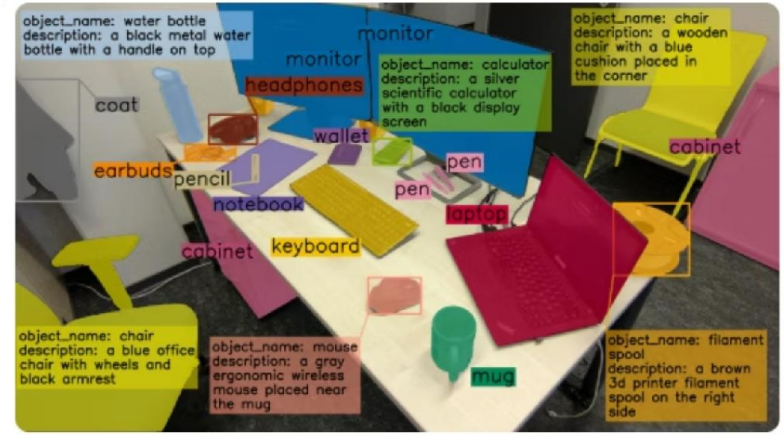
Phase A: Low-Frequency Update

Samples key frames; VLM generates structured JSON object descriptions; MM-Grounding-DINO locates targets with bounding boxes.

Phase B: High-Frequency Tracking

The SAM 2 network translates bounding boxes into precise pixel-level masks, tracked continuously across the video stream with unique Track IDs.

JSON Object File Structure in VLM-GIST



Autonomous Scene Parsing



User-Defined Attributes

This JSON file includes only **object_name** and a **description**, which is limited to **10 words** for efficient filtering and grounding.

```
{
  "object_name": "...",
  "description": "..."
}
```

The fields are **generated by the VLM from the visual scene itself**, without user input. They provide a compact summary before any task-specific attributes are added.

This JSON file extends the autonomous one with user-defined information. It adds additional fields to improve focus and noise filtering.

```
{
  "object_name": "...",
  "description": "...",
  "material": "...",
  "boolean_value": "t/f"
}
```

Unlike the automatically parsed fields, the user-defined fields are **for the downstream task.**

Example Output: Sequential Gaze Events

The system turns raw gaze coordinates into a readable timeline: what was viewed, when, and for how long.

Raw gaze point

Exported from Tobii as time-stamped x,y coordinates.

t = 00:07.8

x = 510 y = 290

gaze point

AOI match

The gaze point is checked against dynamic AOI masks.

Ball

Game

matched to AOI

Readable output

Researchers receive events instead of raw coordinates.

00:07.8 | Table game | Looking at game

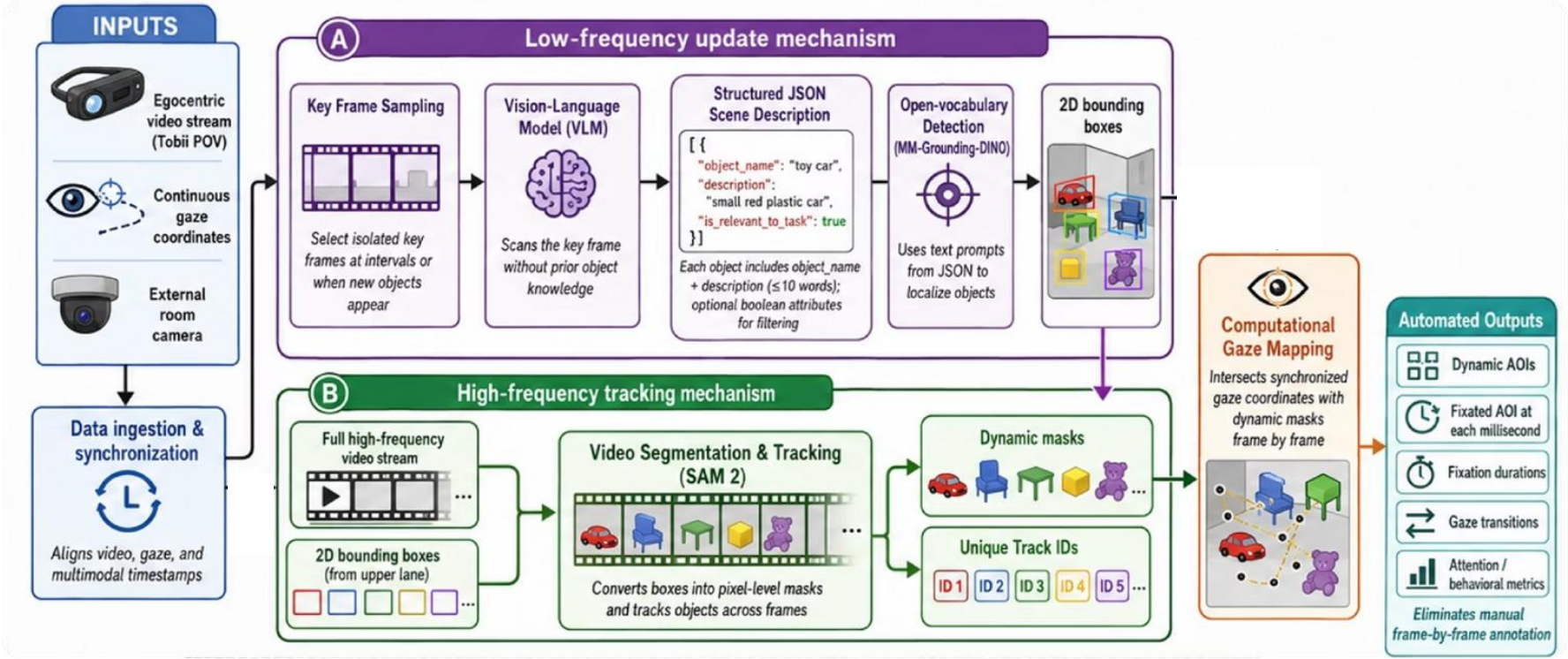
duration: 1.6s

Timeline sample



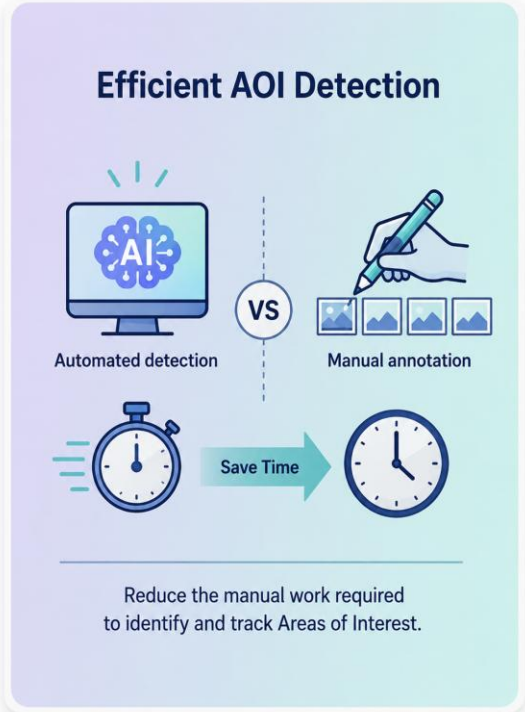
System Architecture

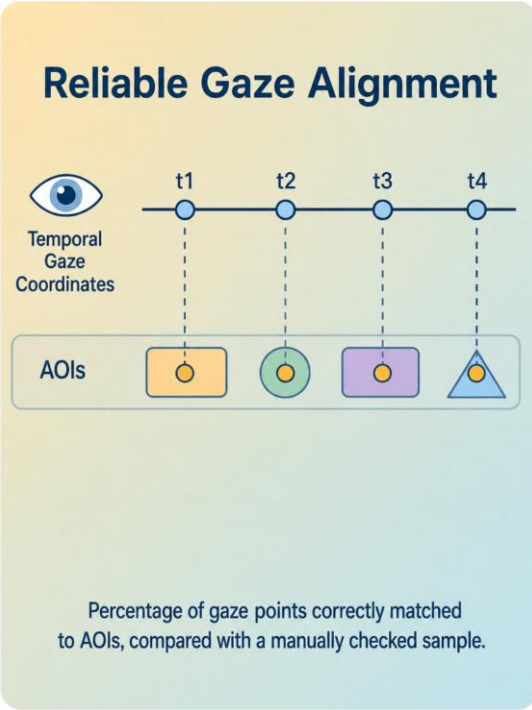
A comprehensive multimodal architecture detailing the data flow from raw egocentric video and gaze coordinates through the tracking mechanisms of VLM-GIST, culminating in automated outputs.



Evaluation Criteria

These criteria outline the metrics and methods which will be used to assess the system performance.





Our Next Steps

01

Main Goal

Determine whether children are visually engaging with the toys they physically manipulate.

02

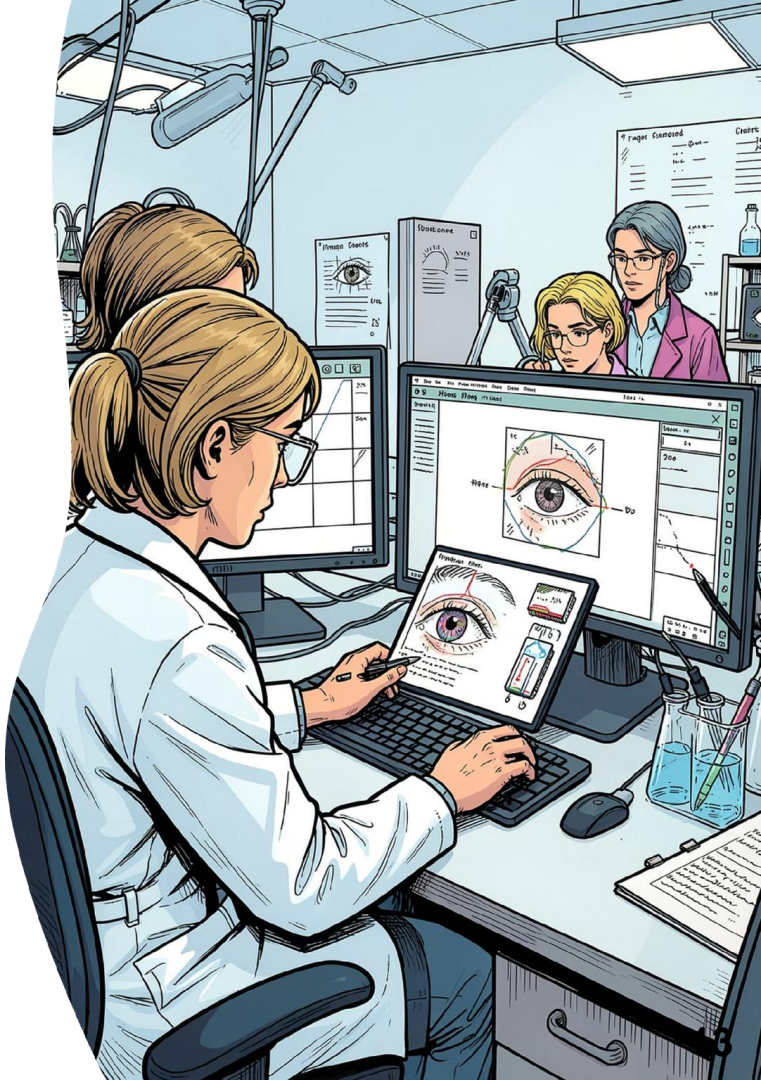
Clear Gaze Events

Connect gaze coordinates to detected objects and extract clear gaze events over time.

03

Multimodal Video Synchronization

Synchronize two different video sources: Tobii POV video and external room cameras for expanded analysis.



Thank You for Listening